

Cepstrum Analysis

BSAI – 7th Semester

Assignment # 3

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Objective

To understand cepstrum analysis and its applications in speech and signal processing, including pitch detection, source-filter separation, echo suppression, and spectral envelope estimation using MATLAB.

Question 1: Cepstrum of an Exponential Signal

Given:

$$x[n] = \left(\frac{1}{2}\right)^n u[n], \quad n = 0, 1, \dots, 63$$

(a) 64-point FFT

The signal is causal and exponentially decaying. Its frequency representation is obtained using FFT.

```
1 n = 0:63;
2 x = (0.5).^n;
3 X = fft(x,64);
```

(b) Real Cepstrum

The real cepstrum is defined as:

$$c[n] = \Re \{ \text{IFFT} (\ln |X[k]|) \}$$

```
1 c = real(ifft(log(abs(X)+eps)));
```

(c) Cepstral Peak and Significance

```
1 stem(c);
2 xlabel('Quefreny');
3 ylabel('Amplitude');
```

Explanation: The dominant peak appears at low quefreny, which represents the exponential decay rate of the signal. This corresponds to the pole of the system in the spectral domain.

Question 2: Complex Cepstrum and Phase Unwrapping

Given:

$$x[n] = \sin(0.2\pi n)u[n], \quad n = 0, \dots, 127$$

(a) FFT and Phase Extraction

```
1 n = 0:127;
2 x = sin(0.2*pi*n);
3 X = fft(x);
4 phaseX = angle(X);
```

(b) Phase Unwrapping

```
1 phase_unwrapped = unwrap(phaseX);
```

(c) Complex Log Spectrum

$$\ln X = \ln |X| + j\angle X$$

```
1 logX = log(abs(X)+eps) + 1j*phase_unwrapped;
```

(d) Complex Cepstrum

```
1 c_complex = ifft(logX);
2 plot(real(c_complex));
3 xlabel('Quefreny');
4 ylabel('Amplitude');
```

Explanation: The complex cepstrum preserves phase information, allowing accurate reconstruction of minimum-phase signals.

Question 3: Source–Filter Separation Using Cepstrum

Given:

$$f[n] = (0.9)^n u[n], \quad h[n] = u[n] - 0.8u[n-1]$$
$$x[n] = f[n] * h[n]$$

(a) Real Cepstrum of $x[n]$

```
1 n = 0:255;
2 f = (0.9).^n;
3 h = [1 -0.8 zeros(1,254)];
4 x = conv(f,h,'same');
5 cx = real(ifft(log(abs(fft(x))+eps))));
```

(b) Lifter Design

Only the first $L_0 = 10$ cepstral coefficients are retained.

```
1 L0 = 10;
2 lifter = zeros(size(cx));
3 lifter(1:L0) = 1;
```

(c) Signal Reconstruction

```
1 cf_hat = cx .* lifter;  
2 F_hat = exp(fft(cf_hat));  
3 f_hat = real(ifft(F_hat));
```

(d) Mean Squared Error

```
1 mse = mean((f - f_hat).^2);
```

Explanation: Low-quefrency components represent the spectral envelope (source), while high-quefrency components correspond to the filter.

Question 4: Cepstrum of a Noisy Vowel

(a) Signal Generation

```
1 fs = 8000;  
2 f0 = 100;  
3 n = 0:1023;  
4 s = sin(2*pi*f0*n/fs);  
5 x = s + 0.05*randn(size(n));  
6 c = real(ifft(log(abs(fft(x))+eps))));  
7 plot(c);
```

(b) Pitch Quefrency

$$T_0 = \frac{1}{100} = 10 \text{ ms} \Rightarrow 80 \text{ samples}$$

(c) Effect of Noise

Noise reduces cepstral peak sharpness and introduces additional small peaks.

Question 5: Minimum-Phase Reconstruction

(a) Minimum-Phase Cepstrum

```
1 cmin = c;  
2 cmin(129:end) = 0;
```

(b) Inverse Cepstral Transform

```
1 Hmin = exp(fft(cmin));  
2 hmin = real(ifft(Hmin));
```

(c) Verification

The impulse response is causal and decaying, confirming stability.

Question 6: Echo Suppression by Homomorphic Filtering

Given:

$$x[n] = s[n] + 0.6s[n - 200]$$

Cepstral Analysis

```
1 cx = real(ifft(log(abs(fft(x))+eps)));
```

The echo delay appears as a peak at quefrency $n = 200$.

Question 7: Spectral Envelope Estimation

```
1 c = real(ifft(log(abs(fft(x))+eps)));  
2 c(21:end) = 0;  
3 env = exp(fft(c));  
4 plot(abs(env));
```

Low-time liftering extracts the vocal tract envelope.

Question 8: Cross-Cepstrum Delay Estimation

```
1 c1 = real(ifft(log(abs(fft(x1))+eps)));  
2 c2 = real(ifft(log(abs(fft(x2))+eps)));  
3 [r,lags] = xcorr(c1,c2);
```

Estimated delay = 50 samples

Question 9: Frame-by-Frame Pitch Tracking

Frames of 20 ms with 50% overlap are used.

Cepstral peaks in each frame give the pitch contour over time.

Question 10: Custom Cepstral Analysis

A musical tone was selected. Cepstral liftering successfully separated harmonic excitation and spectral envelope.

Conclusion

This assignment demonstrated the effectiveness of cepstrum analysis in speech and signal processing. Techniques such as pitch detection, echo suppression, and source-filter separation were implemented and analyzed in detail.